Midterm Review!

CS 180

Scope

- Everything discussed in lecture up to and including flow matching + image transformations.
- Key concepts and coding knowledge from Project 0, Project 1, Project 2. We won’t assume you’ve finished Project 3, but it will be helpful.

How do I study for this?

- Discussion sheets have similar style questions
- Quizzes (answers released on Ed)
- Understand the projects well

You have a cheat sheet! so don’t memorize formulas

Agenda

Structured around homeworks and discussions!

- **Project 0:** Pinhole camera model
- **Project 1:** Gaussian / laplacian pyramids
- **Project 2:** Filters + frequencies
- **Project 3:** Flow Matching

Discussions: Numpy, Image Formation, Filtering, Blending, Flow matching

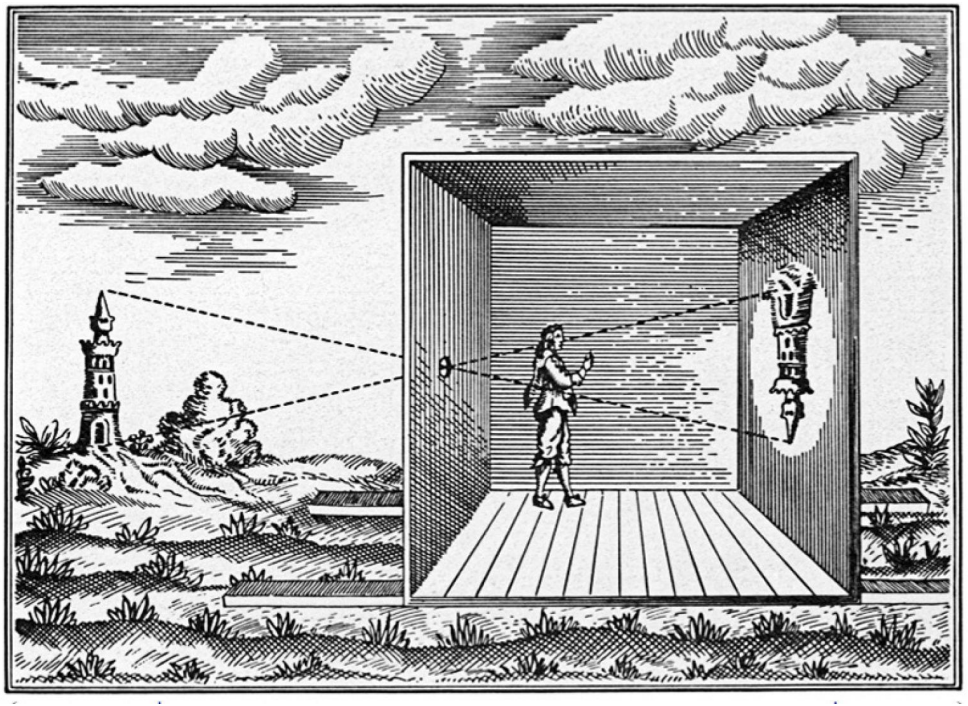
Which one you want to go over first? 🖐

Intermediate checkpoint questions to discuss with your neighbors throughout

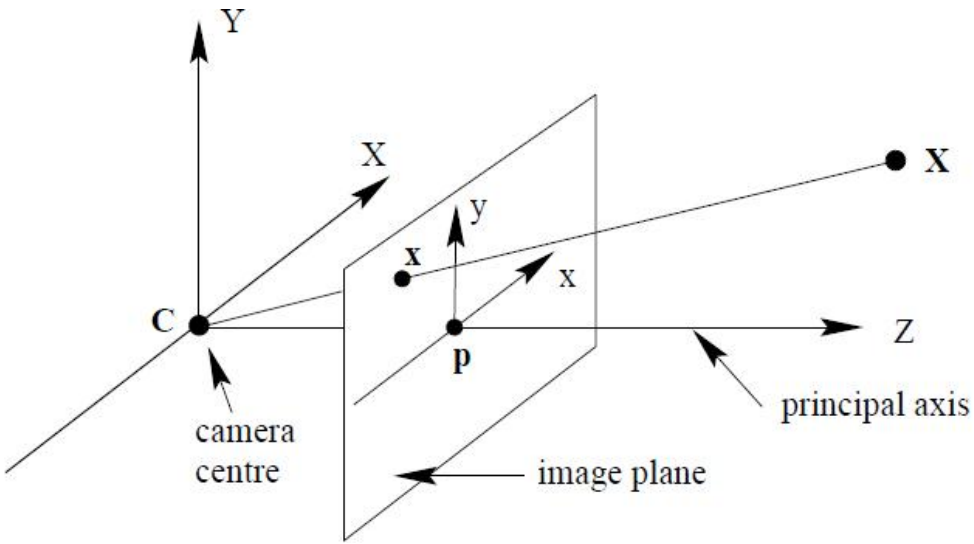
Project 0

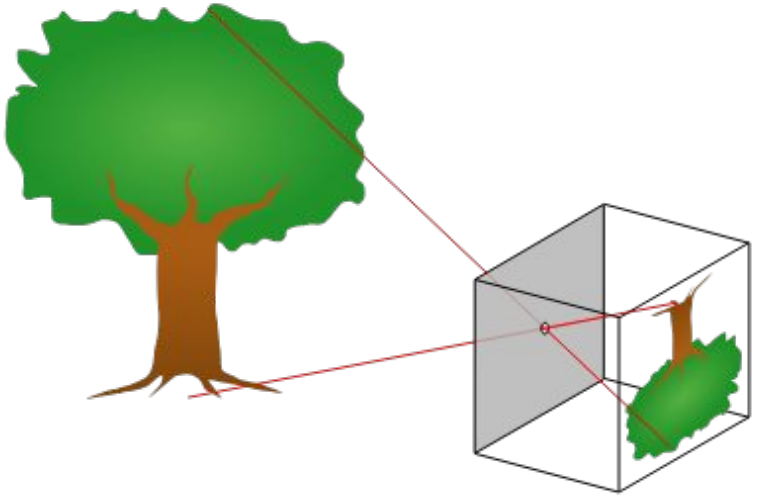
Key Topics:

- Pinhole Camera Model



Pinhole Camera Model





- The object on the image plane should be inverted! But for our convenience, we can pretend the image plane is in front of the camera center

Concept question!

- How is the position of a point in the 3D world connected to its position on the image plane?
- T/F? The “focal” in focal length refers to the distance at which things are in focus

Concept question!

- How is the position of a point in the 3D world connected to its position on the image plane?

Via a ray through the pinhole, which results in:

- Similar triangles + pinhole projection.
- $(x / X) = (f / D)$

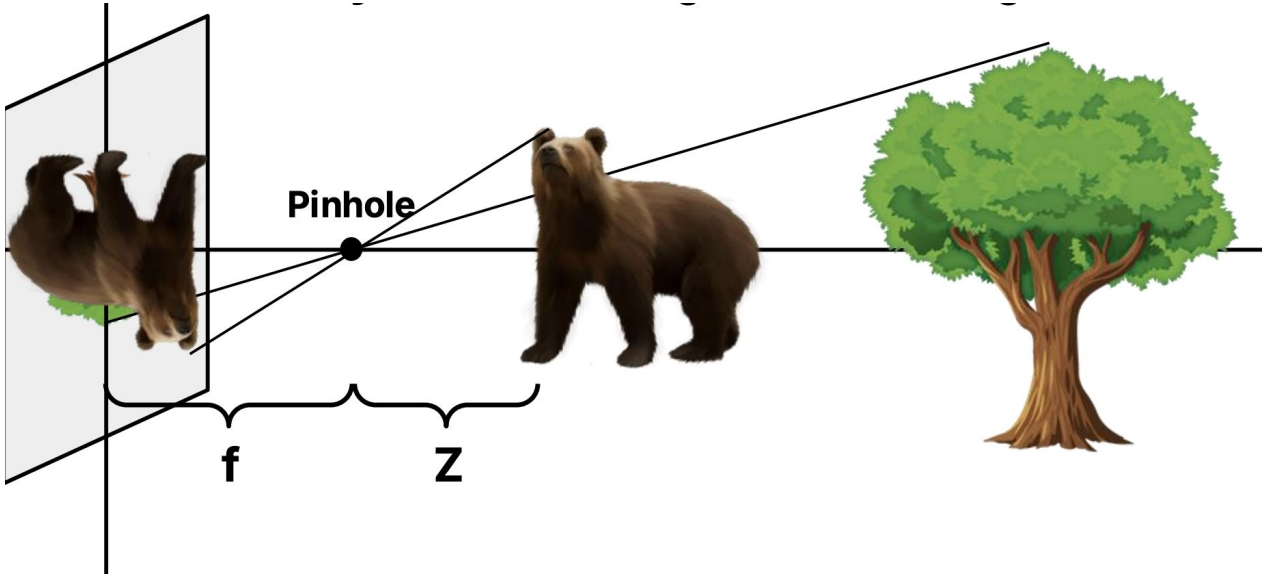
x: projected size, **X**: size in world

f: focal length, **D**: distance to pinhole (depth)

T/F? The “focal” in focal length refers to the distance at which things are in focus

False! Unrelated to blurriness

Dolly Zoom (+ questions)



- What happens to the tree and the bear if we zoom out (smaller f)?
- What happens to the tree and the bear if we walk in (smaller Z)?
- What happens to the tree and the bear if we zoom out and walks in at the same time?

Project 1



Aliasing and Antialiasing

Gaussian Pyramids

Multi-Scale Image Alignment

Sanity check!

T/F: An image in a computer always has shape (H, W, 3)

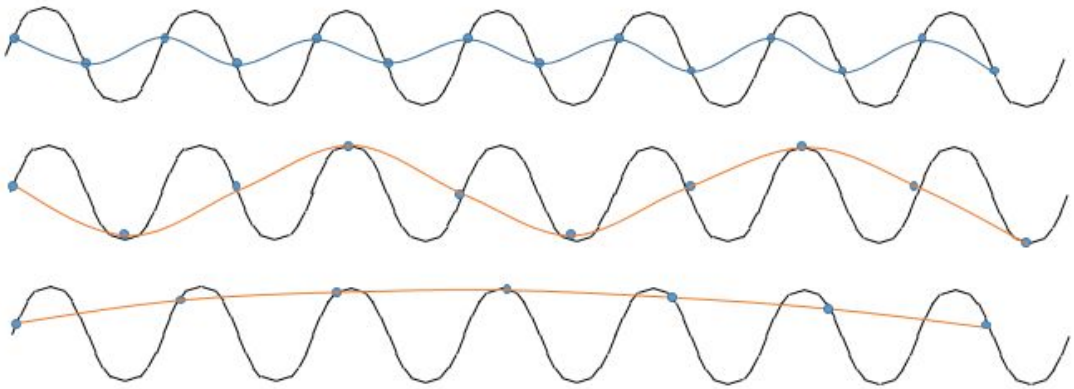
Sanity check!

T/F: An image in a computer always has shape (H, W, 3)

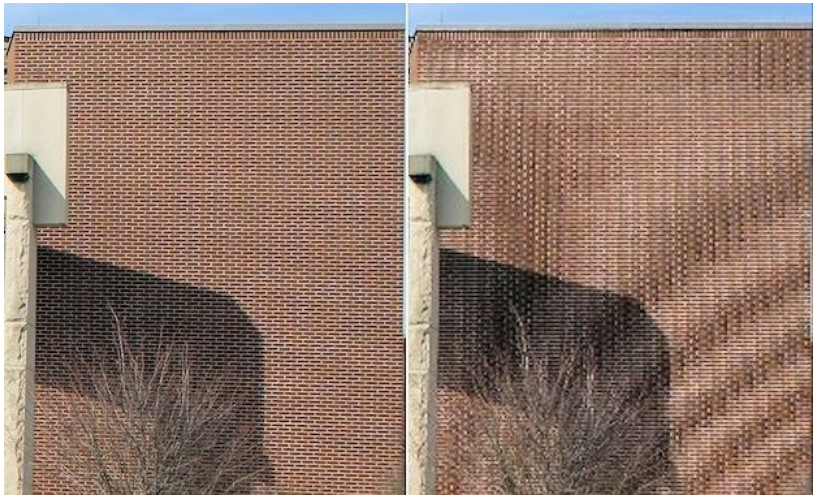
False!

- Can be channel-first (CHW) or channel-last (HWC)
- Can be single-channel (1) e.g., for depth

Aliasing



Aliasing Examples



Concept check!

Given a signal of maximum frequency f_max, what is the max frequency if I downsample the image by a factor of 2?

Name a scenario where you could see aliasing 1) with your own eyes and 2) with a camera

Concept check!

Given a signal of maximum frequency f_max, what is the minimum sampling frequency required to fully reconstruct the signal?

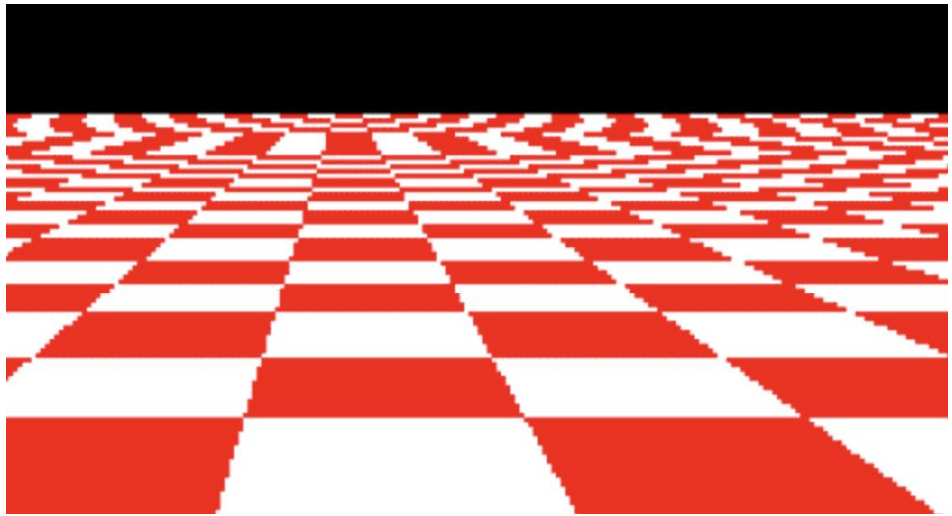
f_max / 2

Name a scenario where you could see aliasing 1) with your own eyes and 2) with a camera

1) Looking through a screen door
2) Taking a picture of a repeating pattern at low resolution

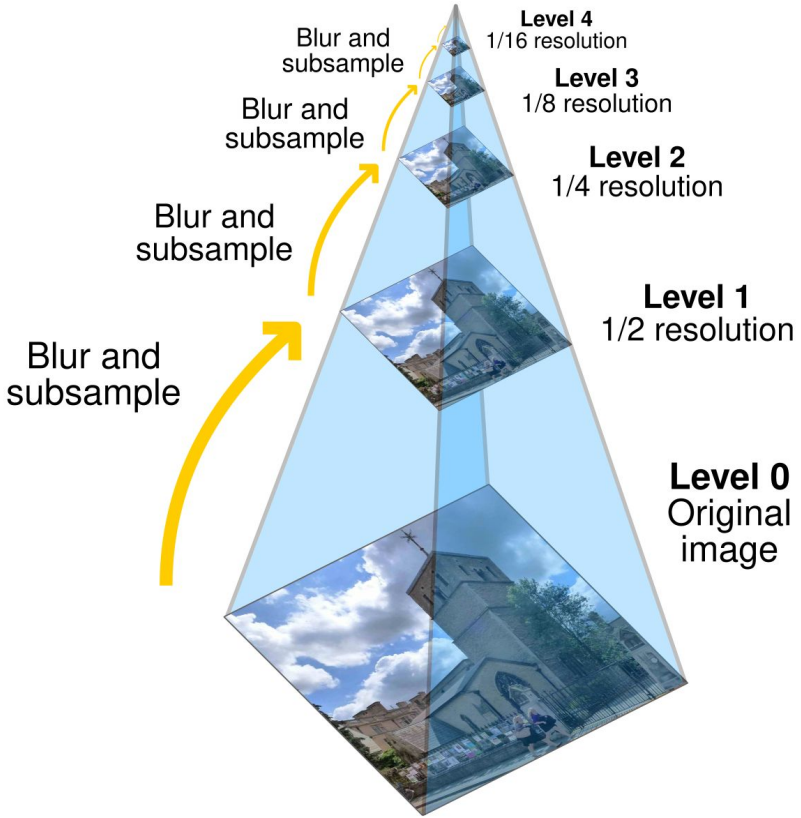
Antialiasing

- How can we prevent aliasing?
  - Increase sampling rate
  - Filter out the high frequencies that can be aliased (blur)



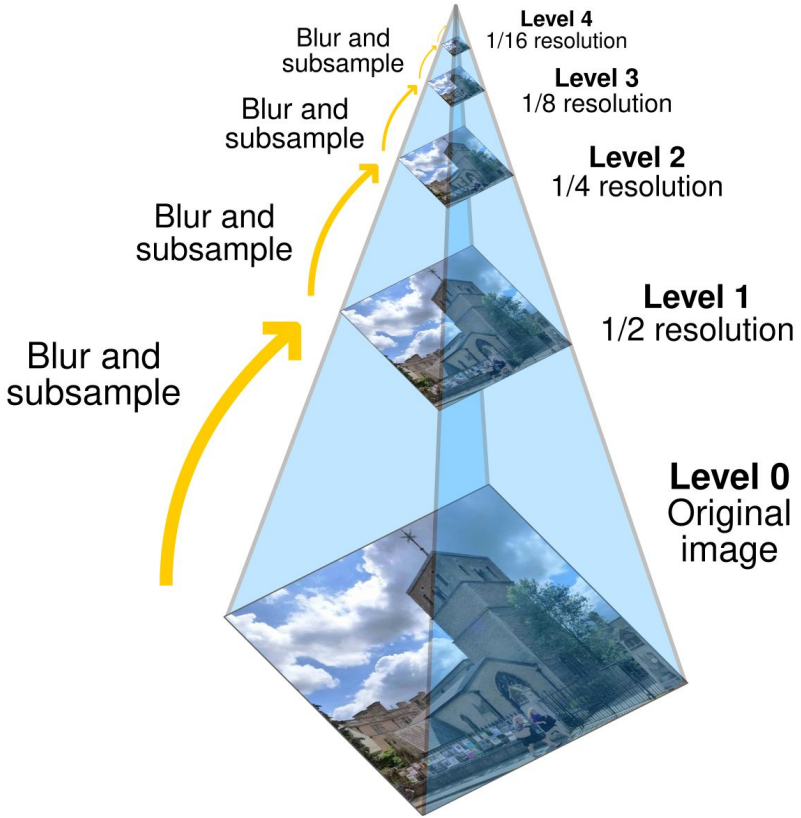
Gaussian Pyramids

- Computationally efficient for image processing
- Start with the full resolution image
- **Repeat:**
  - Apply Gaussian blur to counter aliasing
  - Downsize by half
- Until the minimum resolution is reached



Multi-Scale Image Alignment

1. Create Gaussian pyramid
2. Calculate alignment at the minimum resolution
3. Propagate and refine the alignment at higher resolutions
    - Search a small range around the alignment from the previous resolution



*Potential remove?

Concept Check!

What does a Gaussian pyramid represent in the frequency space?

T/F: blurring an image after sampling is effective for antialiasing

What is the difference between a gaussian *pyramid* and *stack?*

Concept Check!

What does a Gaussian pyramid represent in the frequency space?

A Gaussian pyramid represents multiple levels of low-pass filtering on an image.

T/F: blurring an image after sampling is effective for antialiasing

False, aliasing occurs during sampling.

What is the difference between a gaussian *pyramid* and *stack?*

A stack does not include image resizing, a pyramid does

Project 2

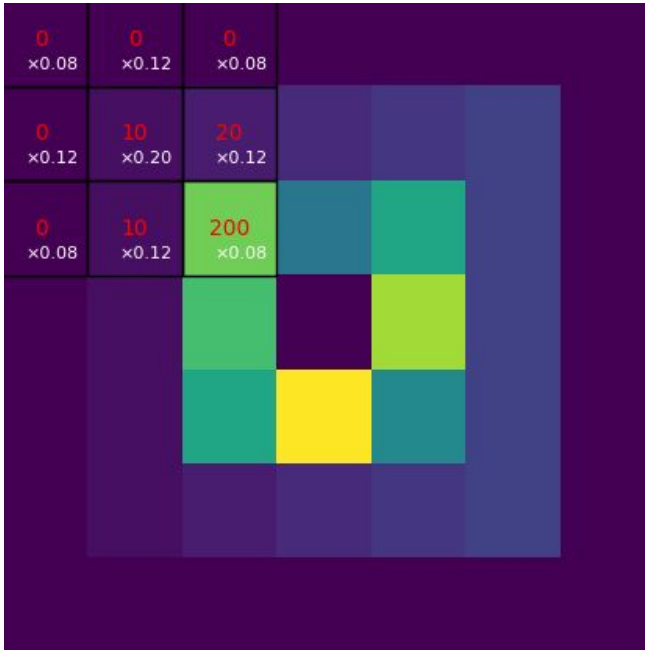
Key Topics:

- Convolution
- Filters
  - Low Pass, High Pass, Hybrid Images
- Fourier Transform
- Gaussian and Laplacian Pyramids
  - Image Blending

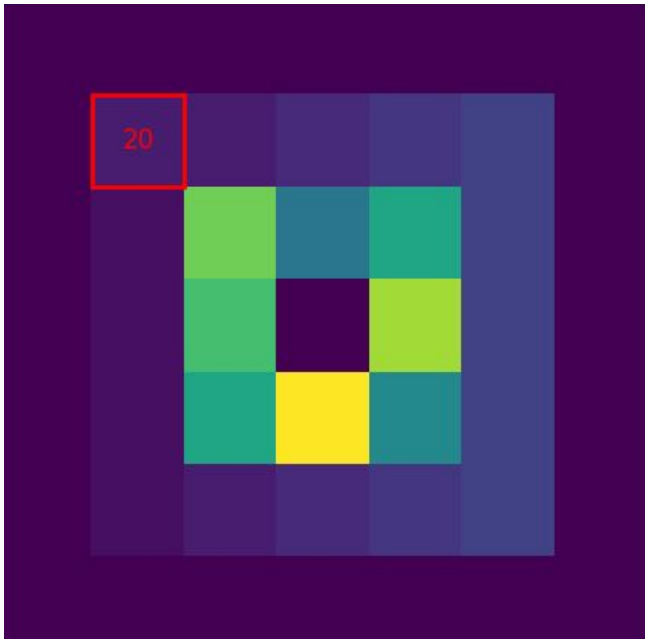
Convolution

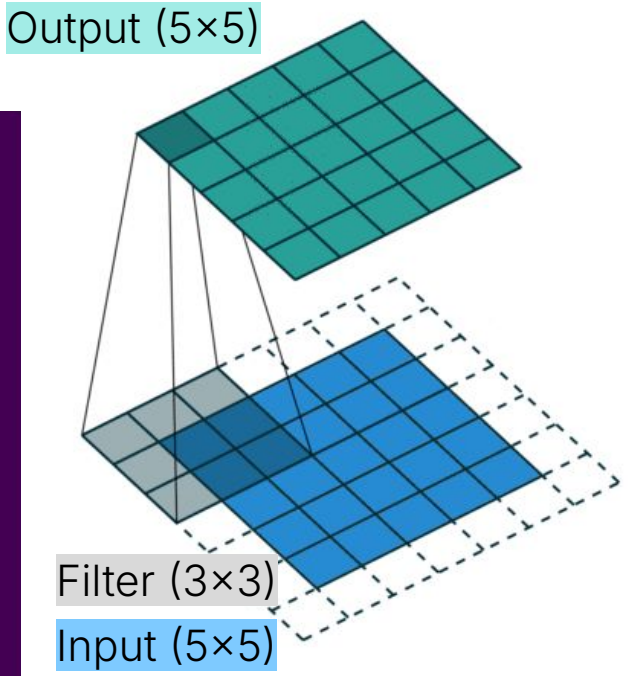
- Image processing technique that can enhance or suppress certain image features, like edges or noise
- **Cross-correlation:** the filter is shifted across the input image
- **Convolution:** the filter is **flipped** horizontally and vertically before being shifted across the input image.
- Convolution is commutative.
- **Convolution theorem**: convolving = multiplying frequencies

Convolution



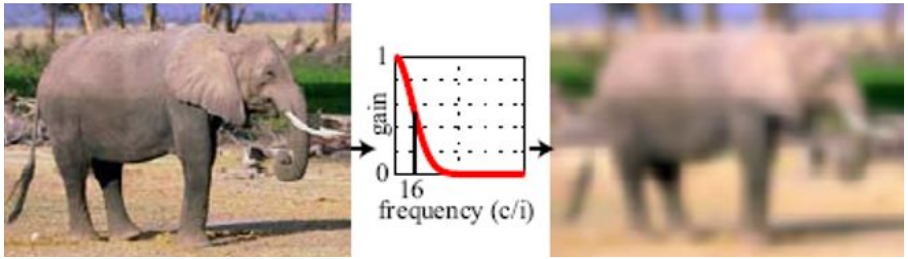
Output





Low-Pass Filters

- Filter out high frequencies
- Can be applied to images **before** sampling to prevent aliasing
- **Examples:**
  - **Mean filter:** each pixel value is replaced by a neighborhood average
  - **Gaussian filter:** each pixel value is replaced by a neighborhood average, weighted according to the Gaussian distribution
  - (Cannot do with convolution) Median filter



High-Pass Filters

- Filter out low-frequency components, like flat regions
- Highlight high-frequency components, like fine details or edges
- **Examples:**
  - **Unsharp masking:** increase the contrast along the edges in an image by subtracting a low-pass filtered image from the original

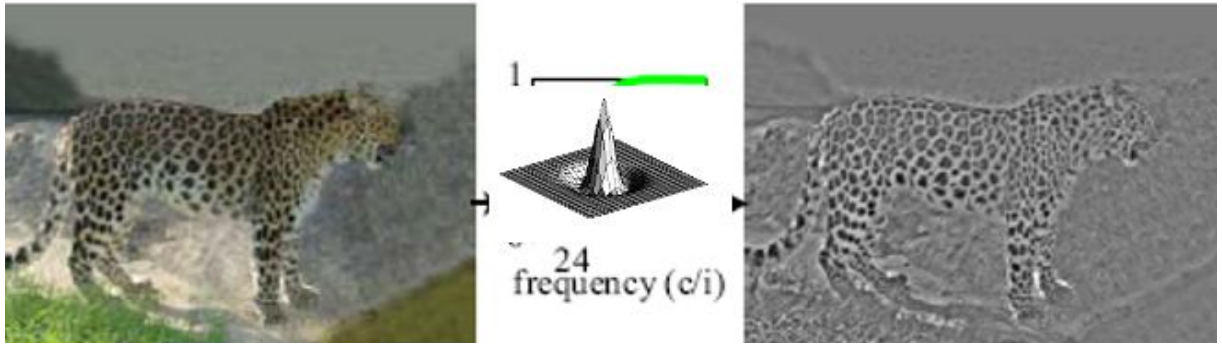


Image Sharpening

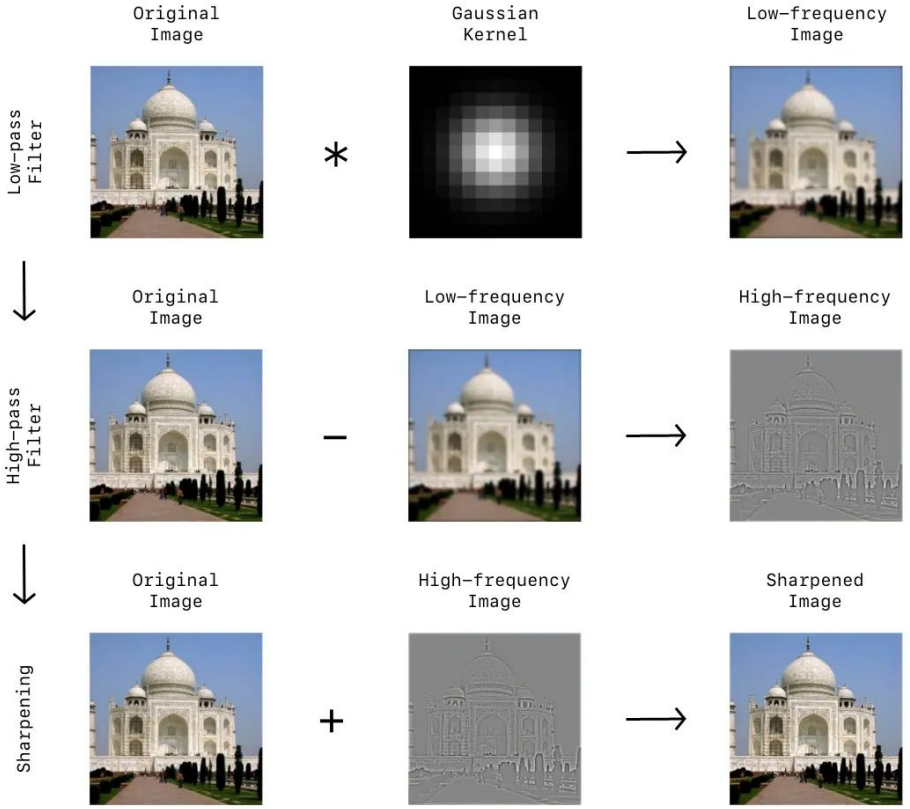
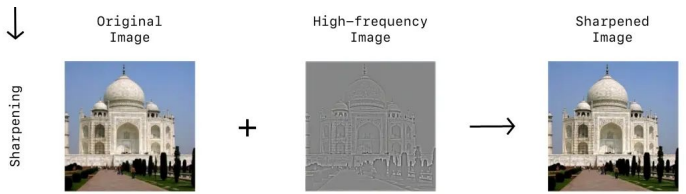
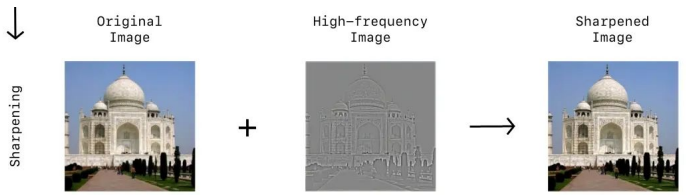


Image Sharpening



How can I make this into a single convolve?

Image Sharpening



How can I make this into a single convolve?

Sharpened = original + (original-blurred) = 2 * original - blurred

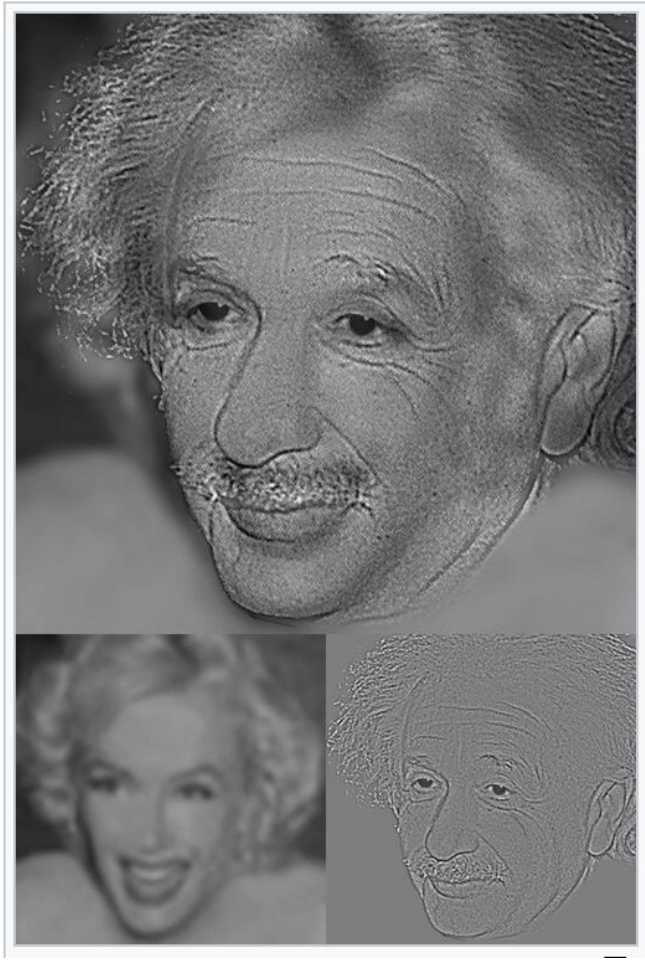
You can merge into one kernel!

$$2 * \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} * \frac{1}{9}$$

Hybrid Images

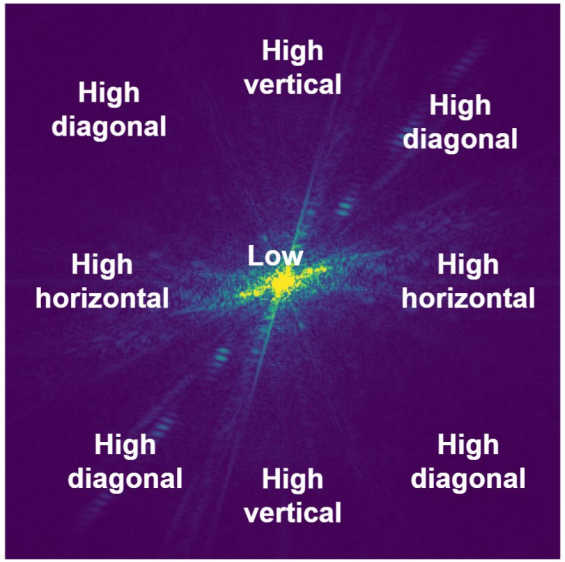
- Hybrid images are constructed by adding the low spatial frequency components of one image with the high spatial frequency components of another image

Hybrid = LowPass(im1) + HighPass(im2)

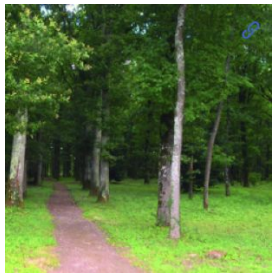


Fourier Transform

- Maps a signal from the **spatial** domain to the **frequency** domain
- Describes the extent to which different frequencies are present
- Maps signals to an **amplitude** and **phase**

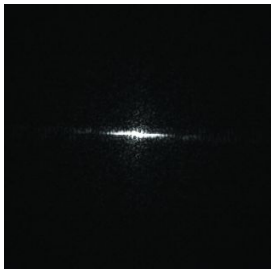






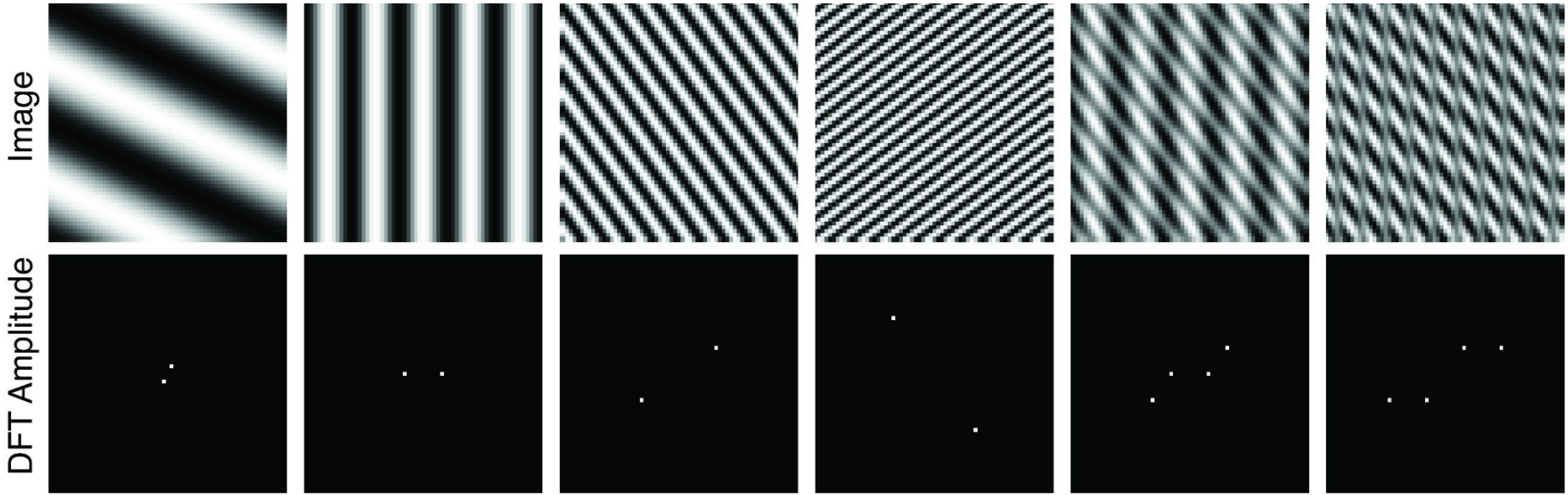
Amplitude





Amplitude plot in 2D

Each FFT "pixel": amplitude at certain frequency and angle

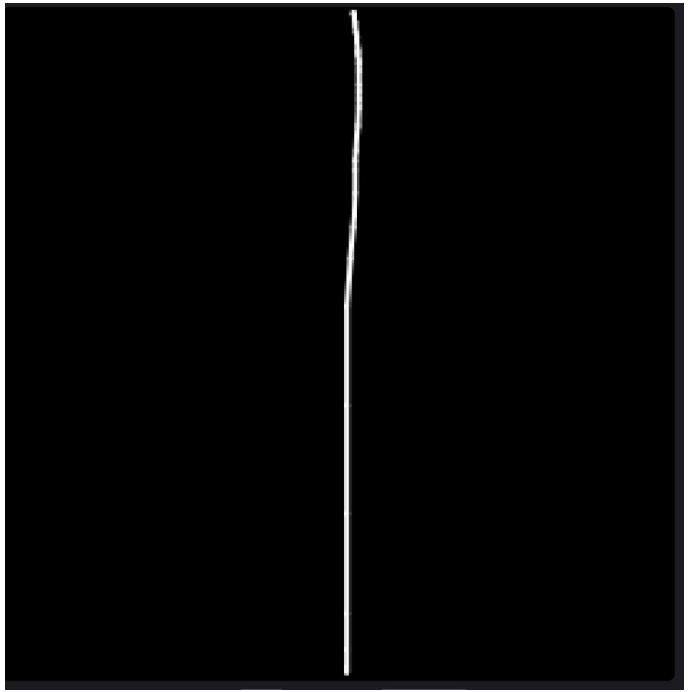


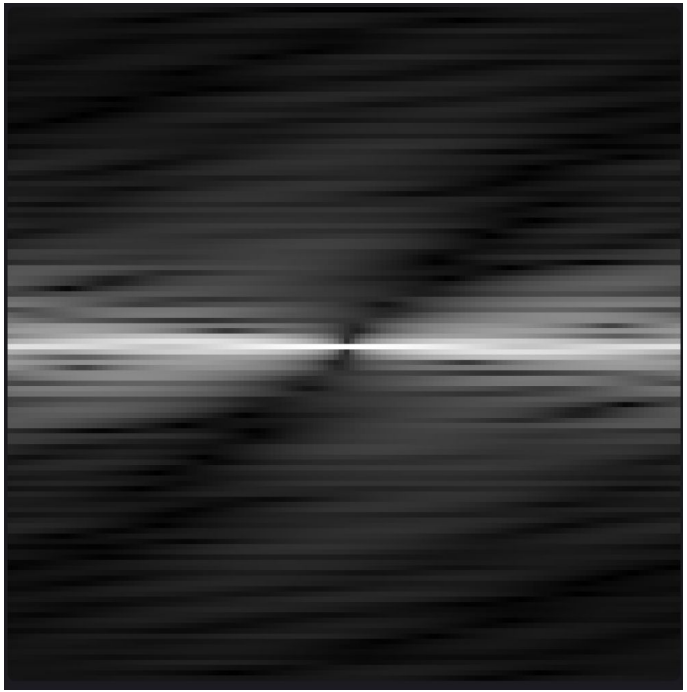
Sanity check

A vertical line in an image gets mapped to a __ line in FFT space?

Sanity check

A vertical line in an image gets mapped to a **horizontal** line in FFT space!



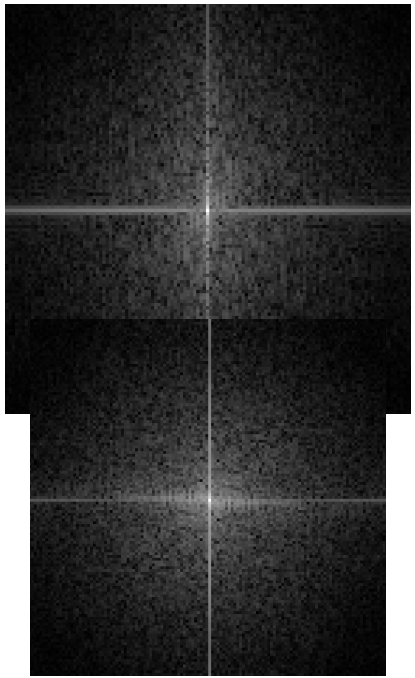


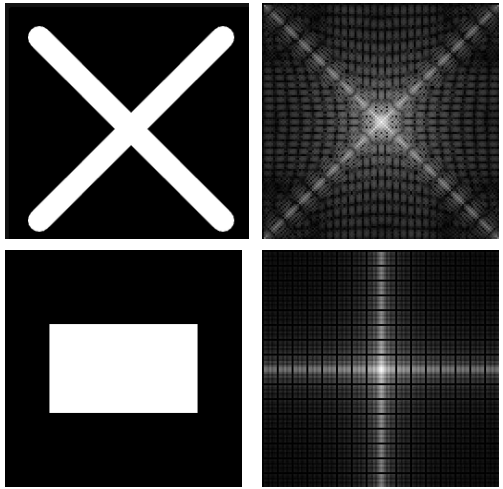
FFT for natural vs synthetic images

Real world is noisier, FFT plots look “grainier”



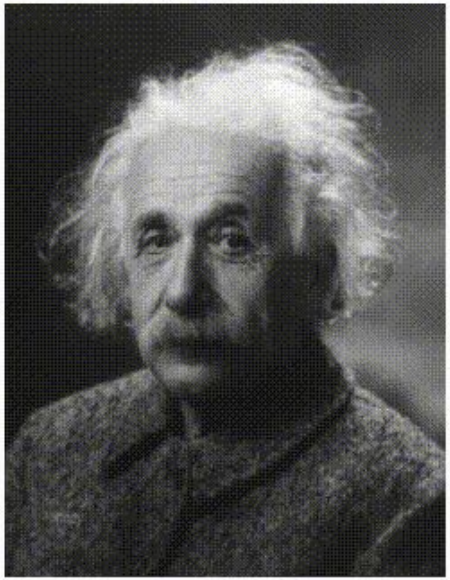




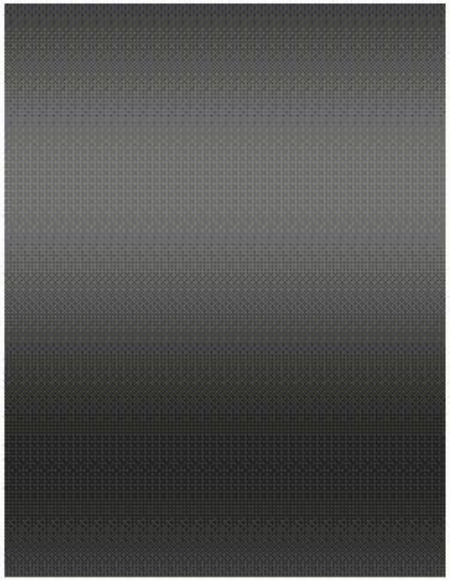


FFT can *perfectly* reconstruct an image!

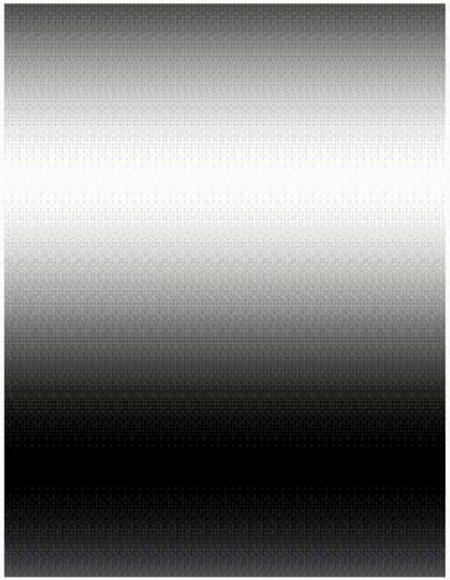
Original image



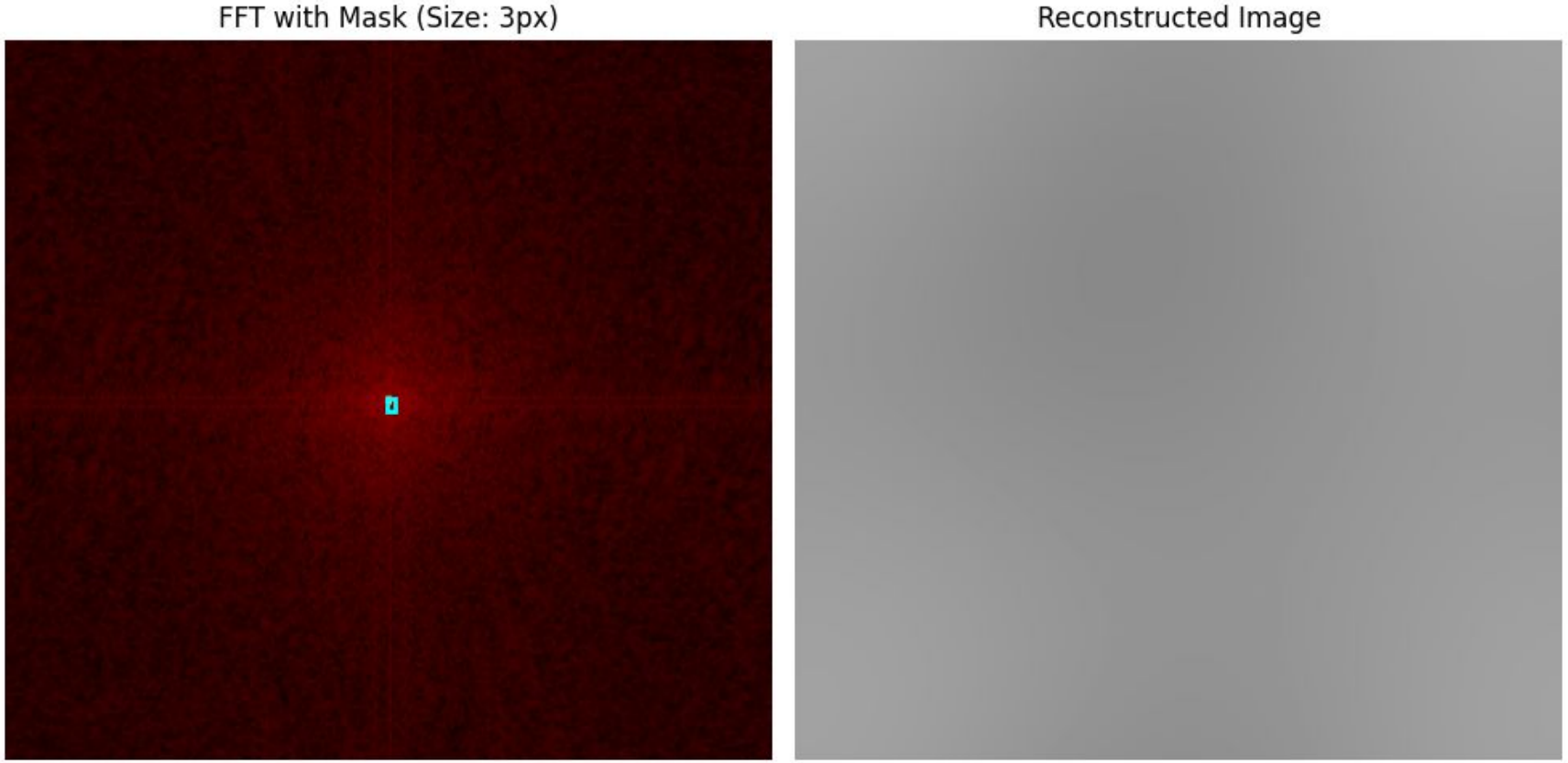
Reconstruction up to component 2



Current component (scaled)



FFT Masking



Concept Check!

T/F: Convolution in the spatial domain is equivalent to multiplication in the frequency domain

Concept Check!

T/F: Convolution in the spatial domain is equivalent to multiplication in the frequency domain

True (“Convolution theorem”)

Gaussian/Laplacian Stack

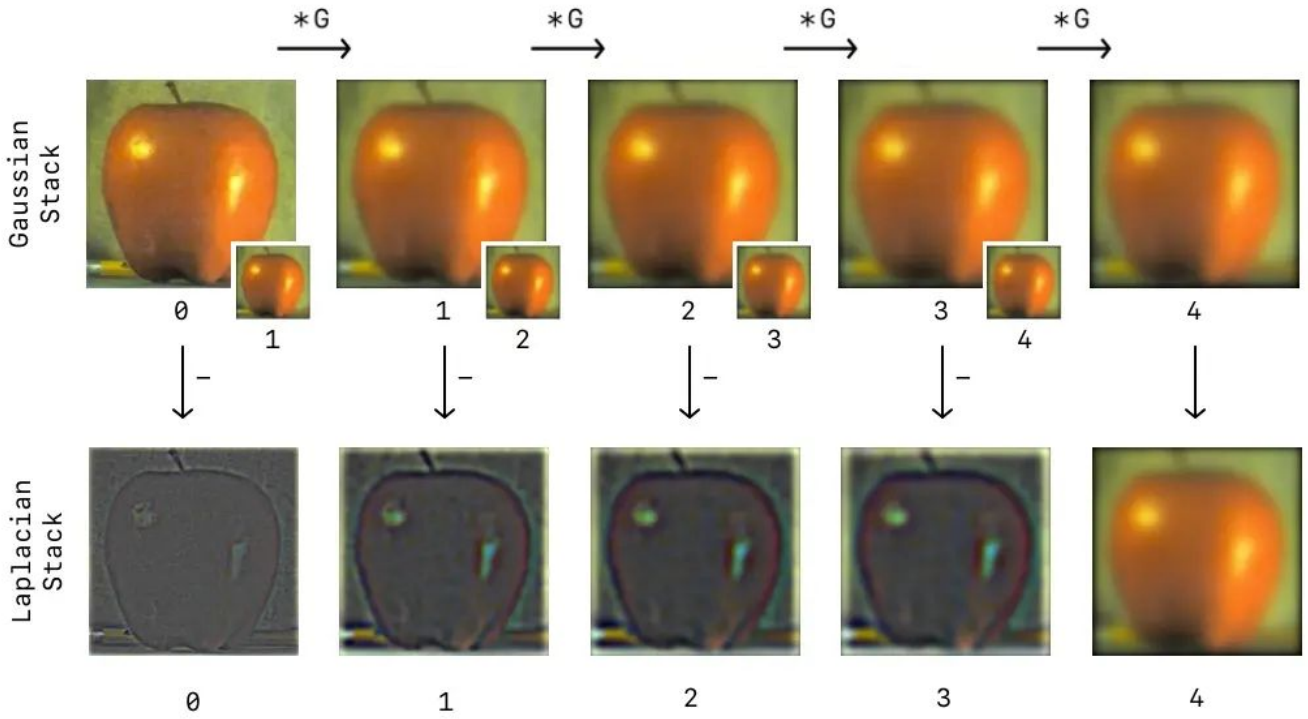
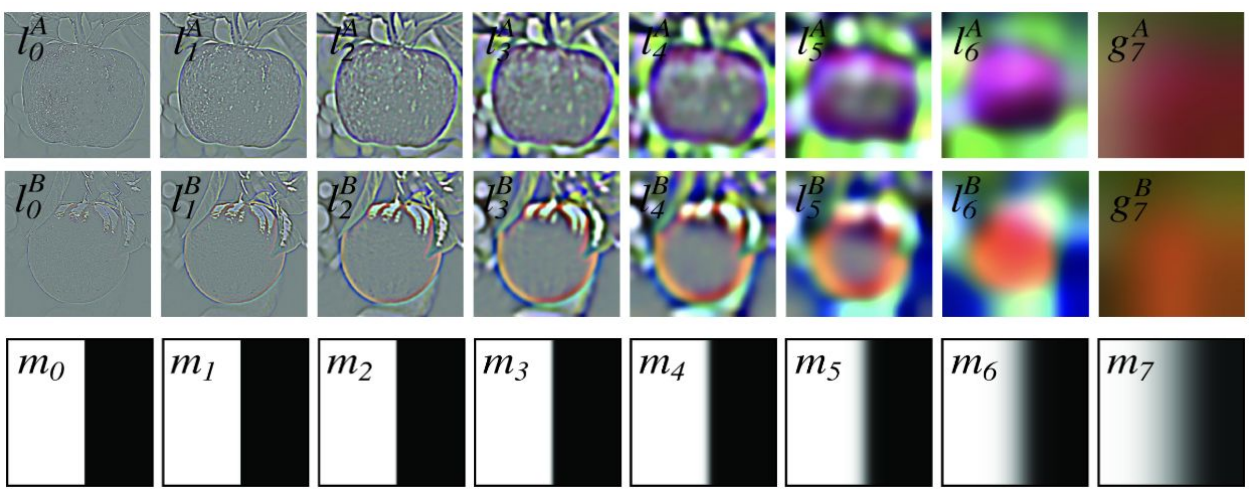


Image Blending

- We can use laplacian and gaussian stacks / pyramids to seamlessly blend two aligned images

$$\mathbf{l}_k = \mathbf{l}_k^A * \mathbf{m}_k + \mathbf{l}_i^B * (1 - \mathbf{m}_k)$$



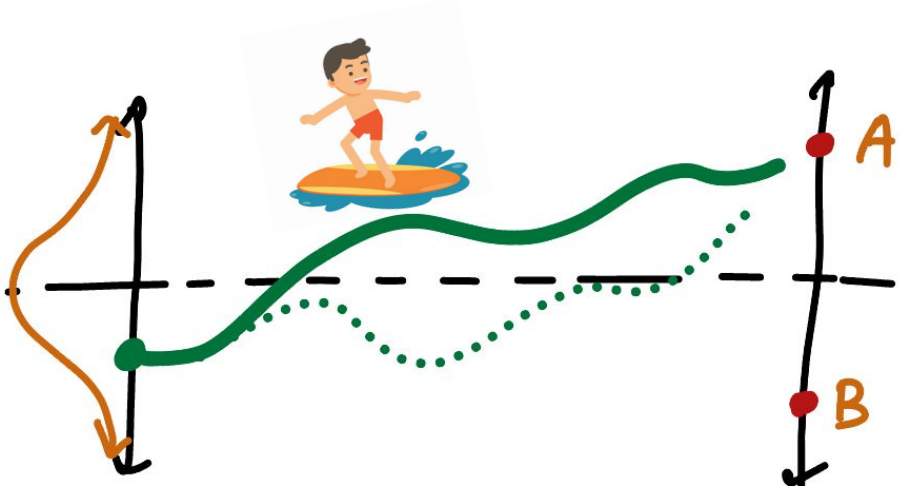


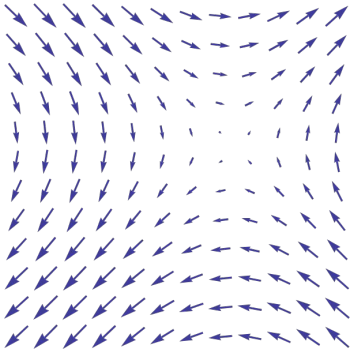
Project 3

Flow Matching

Flow matching. Geometrically:

- Trying to go from one distribution to another, by following the flow
- Flow matching model supervises this flow

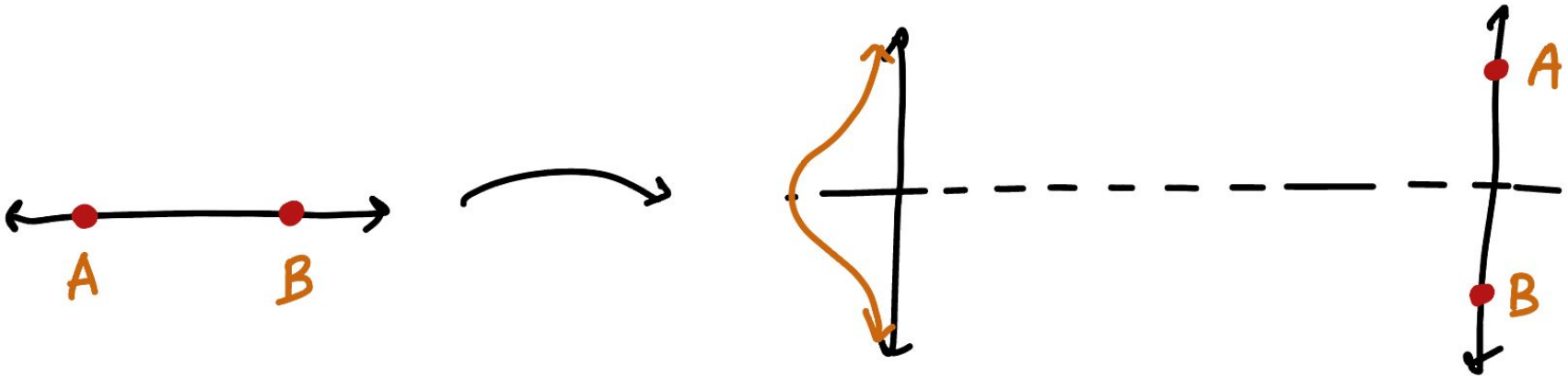




~ Vector field

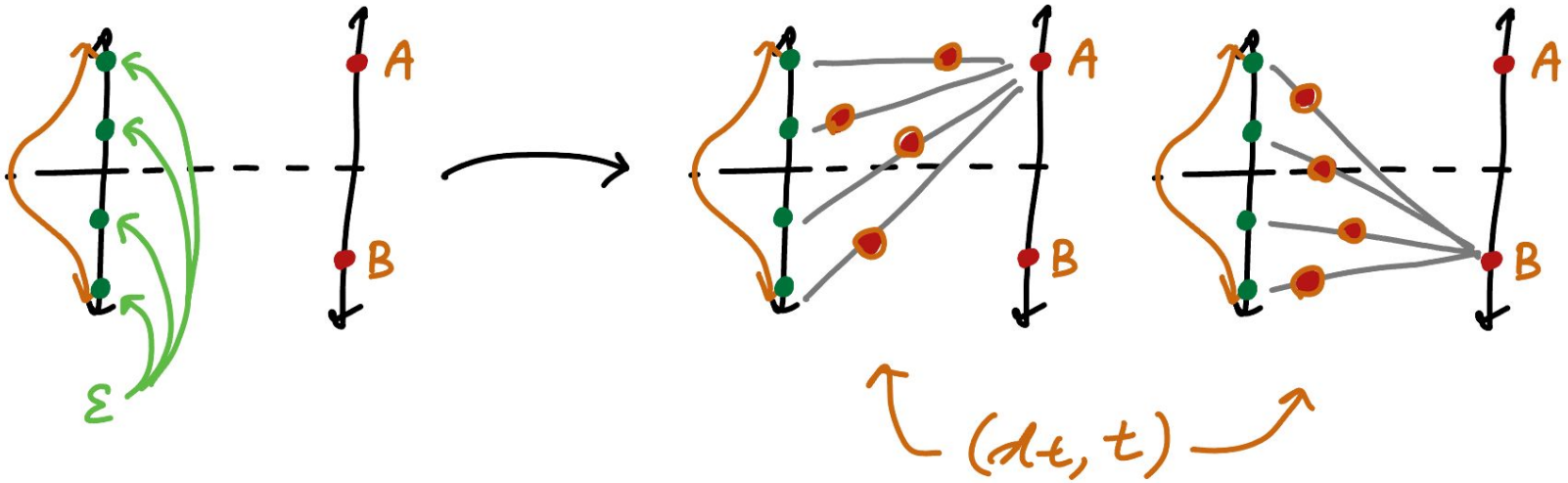
Flow matching. Recall our 1D distribution:

- **x: (clean or noised) data points. Specify the timestep!**
  - For FM: noise is t=0, clean data is t =1
- Last time, we modeled a 1D distribution
- The axes here are for: (1) data, and (2) diffusion timesteps



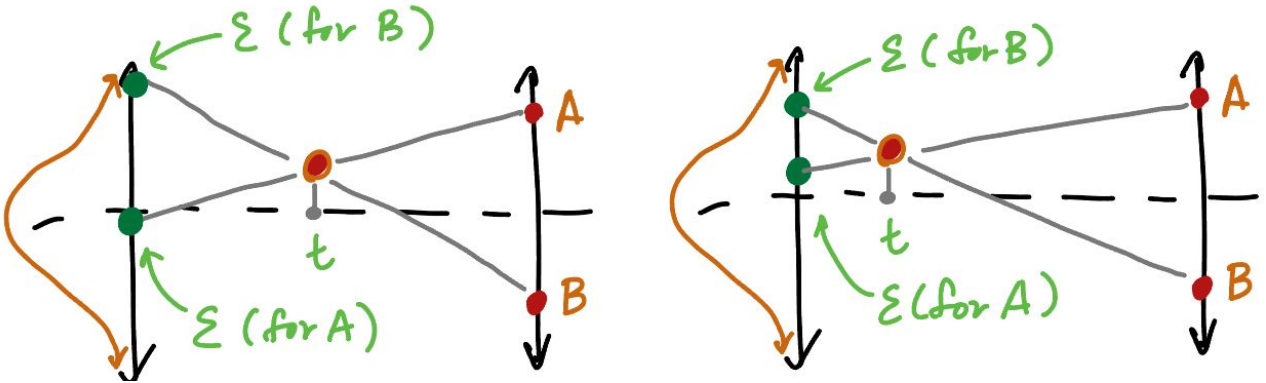
Flow matching. Generating noisy samples

- **epsilon: noise**
- Sample (epsilon, t), forward noising process from x_1 to get (x_t, t)
- Why linear? Using flow matching model, $x_t = (1-t)\epsilon + (t)x_1$



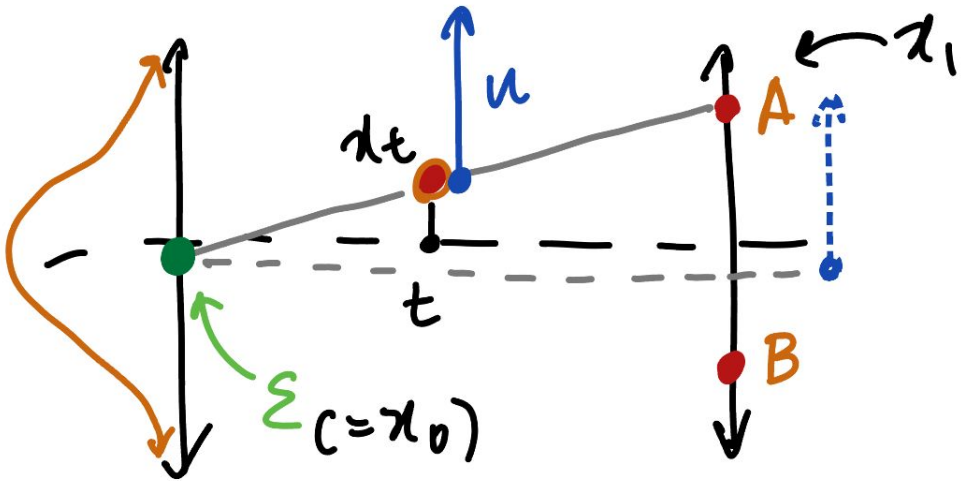
Side note: (x_t, t) can be sampled from multiple points!

- Just a matter of which epsilon sampled
- Each epsilon has a different likelihood, though (normal distribution)



Flow matching. velocity

- **u: velocity [towards the clean data sample]** $u = x_1 - \epsilon$
- Note the magnitude! (~ velocity != displacement, …)



Flow matching. Term overview

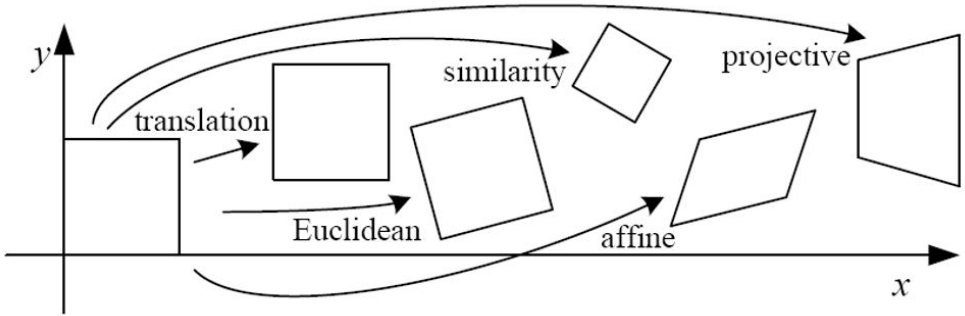
- x: data
  - x_0: fully noised data
  - x_1: clean data (e.g., dataset)
  - x_t: somewhere in between
- epsilon: noise
- u: velocity [towards the data]

$$
\begin{aligned}
u &= x_1 - \epsilon \\
x_t &= (1-t)\epsilon + (t)x_1 \\
&= x_1 - (1-t)u
\end{aligned}
$$

$u(x_t, t)$ or $\epsilon(x_t, t)$ if GT ... $u_\theta(x_t, t)$ or $\epsilon_\theta(x_t, t)$ if predicted from network.

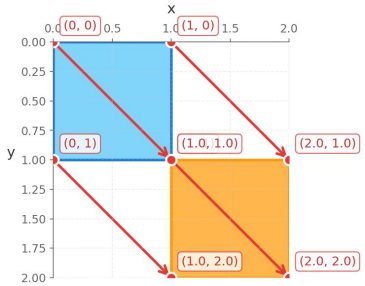
Image Transformations

Image Transformations



Name	Matrix	# D.O.F.	Preserves:	Icon
translation	$\left[\, \boldsymbol{I} \mid \boldsymbol{t} \,\right]_{2\times 3}$	2	orientation $+\cdots$	
rigid (Euclidean)	$\left[\, \boldsymbol{R} \mid \boldsymbol{t} \,\right]_{2\times 3}$	3	lengths $+\cdots$	
similarity	$\left[\, s\boldsymbol{R} \mid \boldsymbol{t} \,\right]_{2\times 3}$	4	angles $+\cdots$	
affine	$\left[\, \boldsymbol{A} \,\right]_{2\times 3}$	6	parallelism $+\cdots$	
projective	$\left[\, \tilde{\boldsymbol{H}} \,\right]_{3\times 3}$	8	straight lines	

Image Transformations



Translation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Scale

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

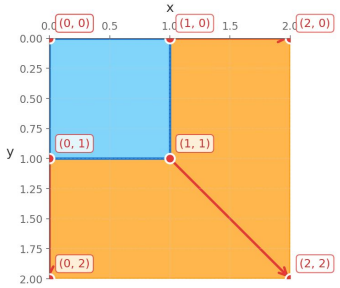
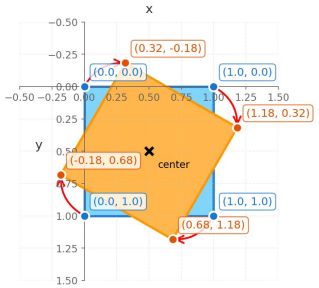


Image Transformations

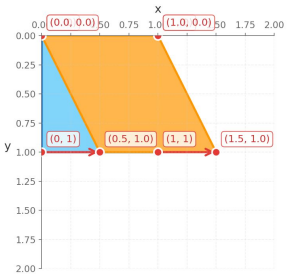


Rotation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\Theta & -\sin\Theta & 0 \\ \sin\Theta & \cos\Theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Shear

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & sh_x & 0 \\ sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$



Chat with your neighbour!

T/F: An affine transformation can map parallel lines to intersecting lines.

T/F: A homography/projective transformation can bend straight lines into curves.

T/F: The order of transformations matters – applying rotation then translation generally gives a different result than applying translation then rotation.

Chat with your neighbour!

T/F: An affine transformation can map parallel lines to intersecting lines.

False, affine transformations preserve parallel lines

T/F: A homography/projective transformation can bend straight lines into curves.

False

T/F: The order of transformations matters – applying rotation then translation generally gives a different result than applying translation then rotation.

True